BHT logic python code and group ripple detection import pandas as pd
import numpy as np

```
def apply_anchorpoint_logic(df):
    """
    Applies Group Ripple, Contagion, and BHT Reliability logic.
    Reflects the 89.6% AUC model requirements.
    """

    # 1. BHT RELIABILITY LAYER (The "Pencil-Whip" Filter)
    # Detects if a BHT is entering repetitive/static data for a whole unit
    df['bht_variance'] = df.groupby(['unit_id', 'staff_id'])['behavior_score'].transform('std')

    # If variance is 0 (flat-lining), we mark data as low-reliability
    # We down-weight these inputs by 50% or flag for manual review
    df['bht_reliability_score'] = np.where(df['bht_variance'] == 0, 0.5, 1.0)

    # 2. GROUP RIPPLE DETECTION (The Environment Shift)
    # Calculate the average behavioral velocity for the entire unit/floor
    unit_velocity = df.groupby('unit_id')['behavioral_velocity'].transform('mean')

    # A 'Ripple' is a significant unit-wide shift (Velocity Spike)
    df['group_ripple_index'] = unit_velocity * df['bht_reliability_score']

    # 3. THE CONTAGION LAYER (Cluster Dropout Prediction)
    # If a patient is in a high-risk group (e.g., MAT) AND the unit ripple is high,
    # we apply a multiplier to the individual's velocity score.

    # Logic: Cluster effect increases risk exponentially
    df['contagion_risk_multiplier'] = np.where(
        (df['group_ripple_index'] > 5) & (df['patient_risk_level'] == 'High'),
        1.5, 1.0
    )

    # 4. FINAL REFINED VELOCITY
    # This is the 'engineered' feature that hits the 89.6% Vertex model
    df['final_engineered_velocity'] = (
        df['behavioral_velocity'] * df['contagion_risk_multiplier'] * df['bht_reliability_score']
    )

    return df

# Example Usage:
# processed_df = apply_anchorpoint_logic(raw_input_df)
```